

TRIDACTYL ADAPTIVEMANIPULATION EVOLUTIONLABORATORY

Simulation-Lab Research Design Framework

TAMEL-1

Document field	Value
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Prepared by	Perplexity AI — Independent Research Design
Requested scope	Optimized three-total-digit manipulators; evolutionary simulation; dexterity; rendering and animation; institutional and worldbuilding applications
Entities addressed	OAL (Omniversal Archive / Architect Labs); ONE INDUSTRIES; LEWIS CORP; simulated environments; THE ONE UNIVERSE; THE MINUS ONE UNIVERSE
Authority status	Independent research proposal; advisory only; not ratified canon, policy, biological conclusion, or product authorization
Source prompt	THREE-DIGITS-RESEARCH.pdf supplied by the user

Core question: Can a morphology with three total digits per hand—including one highly mobile opposable digit—evolve or be optimized to outperform five-digit baselines on defined manipulation and virtual-production objectives?

Document Control

Item	Specification
Research class	Computational morphology–control co-design; virtual embodiment; robotics; generative-media validation
Primary unit of comparison	A complete hand–wrist–controller system, not a five-fingered human hand with two digits removed
Primary claim status	Testable hypothesis. No presumption that three digits are globally superior
Evidence classes	Peer-reviewed research, standards/benchmarks, simulation outcomes, internally governed project materials, and explicitly labeled speculation
Change rule	Any change to morphology constraints, task weights, baseline budgets, or success gates increments the version number and is recorded in a change log
Recommended next version	v1.1 after stakeholder review; v2.0 after pilot results and preregistration

Interpretive Status

This document executes the supplied prompt as an independent research design. It corrects the proposed terminology to “three total digits per hand” or “tridactyl manipulator,” with one digit functioning as an opposable pollex analogue. “Three fingers plus a thumb” would mean four total digits and is outside the primary treatment, although it can be retained as a secondary comparison.

The framework does not attempt to rewrite human evolutionary history, infer that nature seeks simplicity, or claim that fewer anatomical parts automatically produce higher intelligence. It asks an engineering and artificial-life question: under controlled tasks and equalized budgets, which morphology–controller combinations occupy the best regions of a multi-objective performance space?

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Executive Summary

A three-total-digit hand is scientifically and mechanically conceivable as a high-dexterity manipulator, but superiority cannot be inferred from digit count alone. Classic grasp theory shows that three hard-finger contacts with suitable friction and at least three controlled degrees of freedom per finger can, under idealized assumptions, support dexterous object control. The same literature distinguishes manipulative dexterity from grasp robustness: a hand can be excellent at reorienting an object yet comparatively poor at resisting disturbances, power gripping, spanning broad objects, or surviving loss of one contact.

Evidence anchor: Hands for Dexterous Manipulation and Robust Grasping (Bicchi, 2000)

The correct experiment is therefore not “human five fingers versus a human hand after removing two.” It is a morphology–control co-design study. Three-digit candidates should be allowed to compensate through wider opposition, additional abduction/adduction, articulated palms, compliant or variable-stiffness joints, improved tactile sensing, broader pads, different proportions, and redesigned tools. Five-digit and other baselines must receive equivalent optimization effort, actuator or mass budgets, sensing, training compute, and task exposure.

The recommended program combines four workstreams: (1) evolutionary and quality-diversity search over hand morphologies; (2) reinforcement-learning or trajectory-optimization controllers trained with identical curricula; (3) standardized manipulation tests measuring correctness, speed, robustness, energy, complexity, and generalization; and (4) a virtual-production study measuring anatomical consistency, temporal coherence, rigging effort, contact accuracy, and repair burden for three- versus five-digit characters.

The decisive output should be a Pareto atlas rather than a single winner. It should reveal where three digits are best, where five remain best, which compensatory features matter, and whether apparent gains result from true morphology advantages, extra degrees of freedom, favorable task design, or unequal computational budgets. A three-digit system “wins” only if it retains an advantage under resource-matched baselines, unseen tasks, perturbations, and ablations.

The program has direct value for robotics, prosthetics, end-effector design, human–computer interfaces, embodied AI, simulation, animation, generative video, accessibility, and internally governed worldbuilding. Within the named entities, the immediate application is a governed evidence-generating laboratory—not a canon declaration. OAL can preserve the experiment and its scenario branches; ONE INDUSTRIES can evaluate platform and product concepts after its mandate is defined; LEWIS CORP can evaluate real-world-adjacent applications under stricter privacy and safety controls; THE ONE UNIVERSE and THE MINUS ONE UNIVERSE can use validated and counterfactual branches without confusing representation with fact.

Direct Answer

Yes, an optimized tridactyl lifeform or machine could outperform an ordinary five-digit hand on selected dexterity tasks. No, present evidence does not justify predicting universal superiority. The experiment must test morphology, control, sensing, task ecology, and production coherence together.

Research Definition

Primary Research Question

Under equalized physical, sensory, computational, and training budgets, can a three-total-digit manipulator with one opposable digit evolve or be optimized to achieve higher task-weighted dexterity, robustness, and virtual-representation coherence than optimized two-, four-, five-, or six-digit alternatives?

Operational Definition of the Hand

Element	Primary specification	Permitted variation
Digit count	Three total digits per hand	Secondary cohorts: 2, 4, 5, and 6
Opposable digit	One pollex-analogue able to oppose both other digits	Opposition axis, base rotation, length, and force may evolve
Other digits	Two independently actuated digits	Length, phalange count, curvature, lateral motion, and pad shape may evolve
Palm	A load-bearing and contact-capable base	Rigid, cupping, split-palm, or articulated variants
Wrist	Shared reference architecture for the principal comparison	Ablations may test additional wrist freedom
Sensing	Proprioception plus contact/force sensing	Vision and dense tactile arrays introduced by experimental phase
Controller	Same controller family and learning budget across primary baselines	Alternative algorithms used only in robustness studies

Key Terms

Term	Meaning in this program
Dexterity	Ability to change an object’s position or orientation within the hand workspace, complete precise contact-rich tasks, and transition between grasps
Grasp robustness	Ability to retain control despite disturbances, uncertainty, slip, or imperfect object models
Individuation	Ability to command one digit without unwanted movement or force in another
Degrees of freedom (DoF)	Independent kinematic variables available to the hand; digit count and DoF are not interchangeable
Opposition	Ability of one digit to face and contact another digit or palm region
Morphology–control co-design	Joint optimization of physical structure and the policy that controls it
Evolution	A computational search metaphor using selection, variation, and inheritance; not a claim about historical biological evolution
Tridactyl	Possessing three total digits on a hand or terminal manipulator
Virtual coherence	Topological, anatomical, contact, and temporal consistency across poses, views, and frames
Ecological validity	Extent to which the task suite reflects the environment in which the morphology is expected to function

Scientific Rationale

Why Three Is Plausible

Three contacts are mechanically important because they can form a non-collinear support pattern around an object. In a classical hard-finger, nonrolling, nonsliding model, three fingers with three DoF each provide the theoretical nine-DoF minimum described for dexterous manipulation. This does not guarantee performance: friction, geometry, compliance, collision avoidance, sensing error, controller quality, and contact transitions determine whether the theoretical capacity is usable.

Evidence anchor: A Parallel Robots Framework to Study Precision Grasping and Dexterous Manipulation (Borràs et al., 2013)

Human fingers are not perfectly independent. Studies of “enslaving” find unintended adjacent-finger movement and show that independence varies by digit, person, task, and movement range. This supplies a valid design hypothesis: three genuinely independent digits may provide more usable control per digit than five partially coupled digits. It does not establish that fewer digits are globally better.

Evidence anchor: Variable and Asymmetric Range of Enslaving (van den Noort et al., 2016)

Neural control is also not evidently capped at five digits. A case study of two people with well-formed six-fingered hands found independent muscles and nerves, distinct sensorimotor representation, and manipulation abilities unavailable to ordinary five-fingered controls in some tasks. This argues for controller plasticity and against treating five as a neurological maximum; it does not predict which lower-count design is optimal.

Evidence anchor: Augmented Manipulation Ability in Humans with Six-Fingered Hands (Mehring et al., 2019)

Why Evolution Alone Does Not Answer It

Digit reduction has evolved repeatedly in tetrapods through more than one developmental mechanism, including altered early patterning and later tissue remodeling. Those examples demonstrate developmental possibility, not a terrestrial precedent for a three-digit technological species that surpasses humans in manipulation. Most familiar reductions are associated with locomotor or ecological specialization rather than general-purpose precision work.

Evidence anchor: Patterning and Post-patterning Modes of Evolutionary Digit Loss in Mammals (Cooper et al., 2014)

The laboratory must avoid an adaptationist shortcut. Evolution does not search for globally simple or elegant designs; it operates through inherited variation, constraints, local environments, and differential persistence. The simulated result will therefore answer “what emerges under these encoded assumptions and objectives,” not “what nature should have produced.”

Why Simulation Is Appropriate

Morphology–controller co-optimization is already an established robotics strategy. Evolutionary algorithms can search hand geometry and control jointly, while quality-diversity methods can preserve multiple high-performing solutions rather than collapse the study into one weighted score. This is especially useful when dexterity, robustness, mass, energy, manufacturability, and visual coherence conflict.

Evidence anchor: Automated Design of Simple and Robust Manipulators for Dexterous Manipulation

Evidence anchor: MAP-Elites Enables Powerful Stepping Stones and Diversity for Modular Robotics

Objectives and Boundaries

Program Objectives

- › Identify whether optimized three-digit morphologies occupy high-performing regions of a task-diverse dexterity landscape.
- › Separate the effect of digit count from DoF, actuator count, mass, energy, sensing, controller capacity, and training compute.
- › Discover which compensatory features—opposition, lateral motion, palm articulation, compliance, fingertip geometry, or tactile density—make three digits viable.
- › Map task-specific tradeoffs among precision, robustness, power, span, redundancy, multitasking, and damage tolerance.
- › Measure whether purpose-built three-digit character systems reduce rigging, animation, contact, temporal-coherence, and generative-media repair burdens.
- › Produce reproducible artifacts: parameter schemas, morphology archives, controllers, benchmark scenes, logs, videos, and decision records.
- › Generate separately labeled scientific, engineering, production, and worldbuilding conclusions so that speculative findings are not promoted into fact or canon.

Explicit Non-Objectives

- Proving that humans ought to have evolved with fewer fingers.
- Claiming a three-digit system is superior before benchmarks are run.
- Treating a human hand with two digits disabled as an adequate model of an evolved tridactyl hand.
- Optimizing exclusively for one favorable task, such as stylus use or tripod pinching, and generalizing the result to all manipulation.
- Using lower polygon count as a substitute for measuring total production cost.
- Creating biological organisms, editing human embryos, or recommending clinical intervention.
- Assigning canon or operational authority to AI-generated results.

Hypotheses

ID	Hypothesis	Falsifying pattern
H1	Resource-matched three-digit systems can equal or exceed five-digit systems on precision and in-hand orientation tasks.	Five-digit baselines dominate across preregistered precision tasks after equal optimization and uncertainty intervals exclude practical equivalence.
H2	Digit independence, opposition range, wrist mobility, and tactile feedback explain more performance variance than digit count alone.	Digit count remains the dominant predictor after controlling and ablating these variables.
H3	Three digits trade some power, span, redundancy, and multi-object capacity for lower control and production complexity.	No measurable complexity reduction appears, or three digits match/exceed all robustness and redundancy tasks without compensation.
H4	Co-optimizing morphology and control yields better three-digit designs than optimizing either alone.	Fixed-hand learned controllers or hand-designed controllers consistently equal or outperform co-designed systems under equal budgets.
H5	A purpose-built three-digit media pipeline reduces anatomical and temporal hand errors relative to a matched five-digit pipeline.	Digit-count accuracy, contact error, temporal jitter, and human repair time show no material improvement.
H6	General-purpose image/video models retain a five-digit prior that can offset the geometric simplicity of three digits.	Unconditioned models represent three-digit anatomy consistently without specialized references, meshes, or fine-tuning.
H7	No single morphology is globally optimal; winners depend on task ecology and objective weights.	One topology Pareto-dominates all others across every preregistered task and perturbation regime.

Experimental Design

Overall Design

Use a staged, factorial, multi-objective simulation. The primary analysis compares digit count while holding total resource budgets approximately equal. Secondary analyses relax the constraints and allow each topology to find its best physically plausible design. A third analysis coevolves the environment or task distribution to test whether morphology preferences are artifacts of a fixed benchmark.

Comparison Cohorts

Cohort	Purpose	Required controls
2 digits	Lower-complexity pincer baseline	Opposition allowed; equal total actuator and mass budget
3 digits	Primary tridactyl hypothesis	One opposable plus two independently controlled digits
4 digits	Intermediate topology	Same palm envelope and resource constraints
5 digits	Human-like count baseline	Both anthropomorphic and nonanthropomorphic variants
6 digits	Higher-count capacity baseline	Equal budget prevents automatic advantage from extra actuators
Three human digits only	Negative-control ablation	Human geometry with ring/little digits disabled; not treated as an evolved tridactyl hand
Task-specific grippers	Engineering controls	Parallel jaw, suction, and underactuated commercial-style configurations where relevant

Fairness Tracks

Track	What is held constant	Question answered
A: Morphology-only	Controller architecture, training steps, sensors, wrist, and task curriculum	What changes when morphology varies under a common learner?
B: Actuator-matched	Total independently driven axes and peak power	Does finger count matter when controllable resources are equal?
C: Mass/volume-matched	Hand mass, material volume, bounding box, and actuator mass	Is a gain purchased simply by a larger or heavier hand?
D: Energy-matched	Episode energy or integrated torque cost	Which topology produces more useful work per energy budget?
E: Parameter-matched	Policy parameter count and observation bandwidth	Is controller capacity biasing the result?
F: Capability-unconstrained	Each topology receives its optimized design within broad plausibility bounds	What is the best achievable system for each digit count?
G: Production-matched	Equivalent visual fidelity, frame count, camera paths, and task contacts	Does three-digit anatomy reduce total virtual-production effort?

Independent Variables

- Digit count: 2–6 total digits.
- Opposition architecture: fixed thumb analogue; relocatable opposable digit; two-way opposition; radial or central placement.
- Per-digit DoF: flexion/extension, abduction/adduction, axial rotation, base translation, and optional variable-length mechanisms.
- Joint coupling: fully independent, tendon-coupled, underactuated, or learned synergy.
- Palm: rigid, cupping, split, compliant, or actively articulated.
- Fingertip: hard, soft, anisotropic-friction, ridged, deformable, or variable-stiffness pad.
- Sensing: proprioception only; sparse force/contact; dense tactile; vision; visual-tactile fusion.
- Controller: trajectory optimization, model-free reinforcement learning, model-based reinforcement learning, imitation, or hybrid.
- Task ecology: precision-heavy, robustness-heavy, power-heavy, interface-heavy, mixed daily manipulation, or adversarial novelty.
- Production system: skeletal renderer, mesh-conditioned generator, reference-conditioned generator, or general-purpose baseline.

Morphology Genome

Encode each candidate as a constrained graph. Nodes represent palm segments, phalanges, joints, pads, and sensors; edges represent joints, tendons, or compliance links. The genome must support topology changes without producing physically meaningless hands. Hard constraints reject self-intersecting rest poses, disconnected digits, unreachable opposition, negative dimensions, impossible material density, or actuator overlap.

Gene family	Example fields	Constraint
Global	Scale, mass, palm width/depth, wrist pose	Fixed envelope in resource-matched tracks
Digit placement	Base location, angle, opposition role	At least one opposing contact pair
Segment geometry	Length, radius, curvature, taper	Positive dimensions; collision-safe rest configuration
Joint structure	Axis, limits, stiffness, damping, transmission	Plausible ranges; no hidden actuation

Gene family	Example fields	Constraint
Actuation	Torque, speed, power, coupling, tendon routing	Track-specific power/actuator budget
Contact	Pad area, friction model, compliance, texture direction	Randomized friction during robustness tests
Sensing	Sensor count, density, noise, delay	Bandwidth and cost included in objective
Controller interface	Action coordinates, synergy basis, update rate	Matched observation/action accounting

Controller Co-design

Every morphology must receive sufficient adaptation before fitness is assigned. Otherwise, evolution will favor designs that are easy to learn quickly rather than designs with the highest eventual capacity. Use a two-timescale process: inner-loop controller learning for each morphology and outer-loop morphology search. Monitor learning curves and allocate additional inner-loop steps until ranking stability is acceptable.

The primary search should combine a Pareto-based optimizer with a quality-diversity archive. Suggested behavioral descriptors include digit count, total DoF, opposition workspace, hand mass, energy per successful task, robustness score, and controller sample efficiency. MAP-Elites is useful because it preserves the best candidate in many interpretable niches instead of returning only one opaque winner.

Evolutionary Loop

- › Initialize validated morphology families across all digit counts rather than seeding only human-like designs.
- › Train or adapt a controller for each candidate under an identical curriculum and compute budget.
- › Evaluate on training tasks, withheld tasks, perturbations, and production tests.
- › Calculate a vector of performance, robustness, complexity, energy, manufacturability, and coherence metrics.
- › Insert the candidate into a Pareto archive and quality-diversity map if it improves its niche.
- › Mutate geometry, joint architecture, compliance, sensing, and control parameters while preserving feasibility.
- › Periodically re-evaluate archive elites with new random seeds and harder domain randomization to prevent lucky survivors.
- › Stop only when improvement, archive coverage, and ranking stability satisfy preregistered convergence criteria.

Simulation Architecture

System Layers

Layer	Function	Required outputs
1. Morphology generator	Creates validated parameterized hands	Mesh/URDF or equivalent, joints, inertials, collision shapes, sensor map
2. Physics engine	Simulates rigid/soft contact, friction, actuator dynamics, and breakage	State trajectories, contact impulses, energy, penetration, failures
3. Task generator	Samples objects, interfaces, disturbances, and goals	Seeded task manifests and object metadata
4. Controller learner	Trains policies for each candidate	Checkpoints, learning curves, action traces
5. Evolution/QD manager	Selects, mutates, archives, and compares candidates	Lineage, Pareto front, MAP-Elites archive
6. Evaluation harness	Runs blinded train/test and adversarial suites	Metrics with uncertainty and failure labels
7. Media pipeline	Renders skeletal and generated image/video output	Frames, keypoints, contact overlays, repair logs
8. Governance/data layer	Tracks provenance, permissions, status, and review	Immutable run manifest, classification, decision record

Engine Requirements

- Deterministic replay from a run ID, seed, code commit, asset hash, and engine version.
- GPU-parallel simulation for morphology sweeps and reinforcement learning.
- Contact models supporting friction cones, soft pads, restitution, damping, and compliant joints.
- Self-collision, interpenetration detection, actuator saturation, tendon coupling, sensor noise, and latency.
- Optional breakage and digit-disable events for redundancy testing.
- Export to a production DCC/game engine for animation and rendering validation.

- Headless benchmark operation plus visual inspection mode with contact and force overlays.
- Containerized environment and locked evaluation images for reproducibility.

Candidate Technology Stack

A neutral implementation may use MuJoCo, Isaac Lab/Isaac Gym successors, Brax, Bullet, or another validated physics environment; PyTorch or JAX for learning; a quality-diversity library or custom MAP-Elites implementation for search; Blender, Unreal Engine, or a comparable environment for production validation; and a versioned object/task repository. The study should not make engine choice part of the hypothesis. At least one replication should use a second physics engine to expose contact-model dependence.

Domain Randomization

Randomize object mass, center of mass, dimensions, surface friction, compliance, actuator gain, latency, joint backlash, sensor noise, camera position, lighting, and initial pose. Randomization ranges must remain physically credible and must be frozen before confirmatory testing. Fine manipulation research shows that calibration, tactile feedback, and randomized uncertain parameters can materially improve sim-to-real transfer.

Evidence anchor: Dexterous In-Hand Manipulation with Tactile Sensing and Sim-to-Real Transfer

Benchmark Task Ecology

No single task should define dexterity. The suite must cover precision, orientation, stable grasping, power, regrasping, sequential contact, multi-object handling, interfaces, tool use, and failure recovery. Benchmark scores should combine correctness with speed but retain raw metrics so a fast, unreliable hand cannot hide behind an aggregate score.

Evidence anchor: NIST Performance Metrics and Test Methods for Robotic Hands

Evidence anchor: Accessible Open-Source Dexterity Test for Humans and Robots

Task Families

Family	Representative tasks	Primary measurements
Precision placement	Peg insertion; bead placement; connector alignment; stylus path	Success, pose error, contact force, time
In-hand orientation	Rotate cube; roll cylinder; turn irregular object; flip token	Angular range, translation range, drops, finger gait count
Robust grasp	Hold under six-direction disturbances; variable friction; moving base	Maximum resisted wrench, slip, retention probability
Power and span	Lift weighted cylinder; grasp broad irregular object; pull handle	Force, torque, energy, span, tissue/actuator stress
Tool use	Turn screwdriver; operate drill; use scissors analogue; manipulate tongs	Task progress, alignment, recovery, force efficiency
Interface input	Chorded entry; touch surface; rotating node; conventional keyboard control	Correct symbols/minute, error rate, reach, fatigue/energy
Musical sequence	Keyed sequence; pressure–rotation instrument; rhythm reproduction	Timing error, note accuracy, unnecessary contacts
Multi-object	Pick two items; sort clutter; retain one while moving another	Count, selection accuracy, drops, regrasp cost
Fragile/deformable	Handle soft package; grasp thin sheet; transfer fragile object	Damage, peak pressure, slip, completion
Recovery	Digit disabled; sensor dropout; unexpected collision; object geometry shift	Recovery rate, time, residual performance
Novelty	Unseen object category; altered gravity; new interface topology	Generalization gap, adaptation steps
Virtual production	Pose sequence; typing; piano; tool scene; occluded close-up	Topology validity, temporal jitter, contact error, repair time

Curriculum

Stage	Purpose	Examples	Promotion criterion
0. Calibration	Validate physics and joint conventions	Free motion, contact-free reach, single-pad touch	Numerical stability and joint-limit compliance
1. Static grasp	Learn contact acquisition	Spheres, boxes, cylinders	Stable retention without penetration
2. Controlled reorientation	Develop in-hand manipulation	Axis rotations and translations	Pose-error and drop thresholds
3. Contact-rich tools	Develop force regulation	Valves, keys, threaded objects	Task progress under force limits
4. Mixed ecology	Prevent overspecialization	Random daily-object episodes	Balanced family scores
5. Adversarial novelty	Test robustness	Unseen shapes, friction, latency, disabled digit	Bounded performance degradation
6. Production scenes	Test visual coherence	Typing, instrument, assembly, occlusion	Anatomical and temporal acceptance gates

Tool Coevolution Experiments

Run two distinct conditions. In the human-tool condition, every morphology uses fixed human-designed objects and interfaces. In the coevolved-tool condition, handles, controls, key layouts, fasteners, and instruments may adapt alongside the hand. This distinction is essential: a three-digit species should not be judged only by tools optimized for five-fingered humans, but tool adaptation must not be allowed to erase a fair comparison of general manipulation.

- Fixed QWERTY or conventional piano serves as a compatibility test, not the primary ecological benchmark.
- Chorded, pressure-sensitive, radial, spherical, and triangular controls form the coevolved interface set.
- Tool complexity, manufacturability, learnability, and error tolerance are counted so the system cannot “win” with an impossibly elaborate environment.
- Cross-compatibility is measured: how well does each hand operate tools evolved for another topology?

Measurement Framework

Primary Outcomes

Metric group	Measures	Interpretation
Task correctness	Completion, accuracy, pose error, sequence error	Did the hand accomplish the intended task?
Throughput	Correct outcomes per unit time	Combines skill and speed without hiding errors
Dexterous workspace	Reachable object translations and rotations while maintaining control	How much in-hand motion is available?
Robustness	Disturbance tolerance, slip margin, drop probability, parameter sensitivity	Does performance survive uncertainty?
Force and power	Peak grip, sustainable wrench, span, load-to-mass ratio	Can it perform demanding work?
Efficiency	Energy, integrated torque, motion path, actuator effort	What does success cost?
Control complexity	Action dimension, policy size, samples to criterion, inference latency	How difficult is the hand to learn and run?
Morphological complexity	Part count, joints, sensors, actuators, collision pairs, mass	How difficult is the hand to build, rig, and maintain?
Redundancy	Performance after digit/joint/sensor loss	How gracefully does it fail?
Generalization	Held-out object/task/environment performance	Did it learn manipulation rather than memorize scenes?
Production coherence	Digit-count validity, mesh integrity, contact error, temporal keypoint jitter	Is the virtual hand consistently representable?
Human labor	Rigging, cleanup, annotation, and repair minutes	Does simpler topology reduce actual production effort?

Composite Scores

Publish raw measures first. Composite scores may be used for navigation but must not replace the Pareto analysis. One optional score is task throughput: correctness weighted more heavily than normalized speed. A separate robustness-adjusted throughput score multiplies throughput by performance retention under perturbation. Complexity and energy remain separate objectives rather than hidden penalties unless weights are preregistered.

For each task family, report median, interquartile range, bootstrap confidence interval, failure-type distribution, and learning curve. Also report worst-decile performance, because a hand that is excellent on average but catastrophically unreliable may be unsuitable for tools, accessibility, or safety-sensitive applications.

Finger Independence Index

Command one digit through its range while the others are instructed to remain still. Define individuation as one minus the normalized unintended movement or force of non-commanded digits. Calculate the measure separately for small and large movements and for active versus passive coupling. Do not award full independence merely because controller commands are separate; use observed kinematics and forces.

Virtual Coherence Index

Component	Example measurement
Digit count accuracy	Fraction of frames containing exactly the specified visible and inferred anatomy
Topology integrity	No fused, duplicated, detached, or discontinuous digit geometry
Temporal stability	Frame-to-frame keypoint acceleration after removing intended motion
Contact validity	Fingertip/key or fingertip/object distance and penetration at intended contact time
Occlusion recovery	Correct anatomy before and after self-occlusion
View consistency	Same anatomy across synchronized camera angles
Semantic task fidelity	Correct key, note, symbol, or tool action occurs
Repair burden	Human minutes and regeneration attempts needed for acceptance

Detector confidence may be recorded but cannot be the only anatomical metric, especially for a novel three-digit species that ordinary human-hand detectors were never trained to recognize. Use a species-specific keypoint/mesh validator plus blinded human review and task-level contact checks.

Evidence anchor: HandSurge: 3D Mesh-Guided Correction of Generated Hands in Video

Analysis Plan

Confirmatory Model

For each primary outcome, fit a hierarchical model with fixed effects for digit count, total DoF, individuation, opposition workspace, compliance, sensing, task family, and relevant interactions. Include random effects for evolutionary seed, controller seed, object instance, and scene. Repeated episodes from the same morphology must not be treated as independent samples.

The central contrast is optimized three digits versus optimized five digits under the actuator-, mass-, energy-, and training-matched tracks. Equivalence margins must be established during the pilot using practical task tolerances. Report superiority, inferiority, or practical equivalence rather than relying only on whether a p-value crosses a threshold.

Multi-objective Analysis

- Plot Pareto fronts for dexterity, robustness, energy, morphology complexity, control complexity, and production coherence.
- Compute archive coverage and quality-diversity score for the morphology landscape.
- Perform sensitivity analysis across alternative task weights rather than selecting one convenient weighting.
- Identify crossover points—for example, the object span or disturbance magnitude at which five digits overtake three.
- Use Shapley-style or permutation attribution cautiously to estimate which design variables explain performance; confirm with ablations.
- Retain failed and low-performing candidates to avoid survivorship bias in the morphology archive.

Ablations

Ablation	Question
Three digits with human-like joints only	Does topology alone help without enhanced articulation?
Three digits without tactile sensing	Is tactile compensation essential?
Three digits with coupled tendons	Does digit independence drive the benefit?
Rigid versus articulated palm	Can palm morphology recover power and span?
Fixed versus expanded wrist	Is apparent hand dexterity actually wrist motion?
Equal DoF versus equal digits	Is the advantage caused by DoF allocation rather than count?
Fixed controller versus co-designed controller	Does optimization method create the result?
Fixed tools versus coevolved tools	Does the environment favor one anatomy?
Mesh-guided versus prompt-only generation	Is visual consistency due to explicit structure?
Matched polygon count versus matched visual quality	Where do actual production savings arise?

Replication and Sample Planning

Use a pilot of at least 10 independent seeds per major topology–controller condition to estimate variance, compute demands, failure modes, and practical effect sizes. The confirmatory phase should target at least 30 independent evolutionary seeds for the primary three-versus-five comparisons, with the final count determined by a preregistered simulation-based power analysis. Every seed receives the same maximum environment interactions and wall-clock accounting. Additional episodes within a seed improve estimate precision but do not replace independent seeds.

A blinded evaluation server should load frozen candidate checkpoints and run withheld tasks. Developers may inspect training metrics but should not see confirmatory object identities, exact perturbation schedules, or final task weights until submission is locked.

Generative Media Workstream

Question

Does a purpose-built three-digit anatomy measurably reduce total representation and animation complexity compared with a matched five-digit anatomy, and under what generation method? The workstream distinguishes traditional rigged rendering from learned image/video generation. Raw triangle count is expected to be a small part of the cost; rigging, skinning, self-occlusion, collision/contact, temporal tracking, model conditioning, and quality assurance are more plausible savings channels.

Conditions

Condition	Three-digit version	Five-digit version
Rigged 3D	Matched materials and visual fidelity; three-digit skeleton	Matched materials and visual fidelity; five-digit skeleton
Prompt-only image	Text requests anatomy without structural guide	Text requests ordinary human-like hand
Pose-conditioned image	Species-specific 2D/3D keypoints	Human-hand keypoints
Mesh-conditioned video	Explicit three-digit mesh across frames	Explicit five-digit mesh across frames
Fine-tuned model	Training corpus balanced for tridactyl anatomy	Comparable corpus for five-digit anatomy
Hybrid production	Generate, then rig/mesh-guided repair	Generate, then rig/mesh-guided repair

Scene Set

- Neutral open-hand turntable with front, side, oblique, and close-up views.
- Self-occluding grasp sequences with rapid finger crossings.
- Typing on a conventional keyboard and on a topology-adapted chorded interface.
- Playing a keyed instrument with exact note ground truth.
- Rotating a translucent object so contacts remain visible.
- Using a tool requiring continuous force and intermittent regrasp.
- Two-hand cooperation and hand-to-hand object transfer.
- Low light, motion blur, reflective materials, gloves, and partial occlusion stress tests.

Production Metrics

- Asset creation hours: modeling, topology, rigging, weight painting, corrective shapes, and test poses.
- Animation hours: blocking, contact correction, finger collision cleanup, and retargeting.
- Runtime: skinning cost, contact-solver time, inference time, memory, and output frame time.
- Quality: anatomical validity, keypoint jitter, mesh penetration, contact correctness, and task fidelity.
- Repair cost: regenerations, inpainting passes, artist interventions, and rejected shots.
- Consistency: stable identity and digit topology across camera changes and scene cuts.

Generated hands remain a known failure area, and mesh-guided refinement methods explicitly target malformed anatomy and depth ambiguity. The test must therefore compare general models with purpose-built conditioning rather than assuming the generator internally counts fingers or maintains a stable skeletal representation.

Evidence anchor: HandRefiner: Refining Malformed Hands in Generated Images

Validity and Failure Controls

Threat	How it could mislead	Control
Task bias	Tripod-friendly tasks predetermine a three-digit win	Balanced task ecology; publish family-level results
Resource bias	More DoF, power, sensors, or compute assigned to one topology	Multiple matched-budget tracks and full accounting
Learning-speed bias	Fast-learning morphologies are evaluated before slower designs mature	Ranking-stability tests and adaptive inner-loop training
Physics bias	One engine's friction/contact approximations favor a morphology	Second-engine replication and parameter sweeps
Tool bias	Human tools favor five digits; coevolved tools favor the morphology that designed them	Report fixed-tool and coevolved-tool results separately
Metric collapse	One composite score hides catastrophic weakness	Raw metrics, Pareto fronts, and worst-decile outcomes
Overfitting	Morphologies memorize objects or task sequences	Held-out categories, procedural objects, blinded server
Detector bias	Human-hand detector penalizes valid tridactyl anatomy	Species-specific validator plus human and contact review
Sim-to-real gap	Ideal simulation exaggerates contact skill	Noise, latency, backlash, calibration, tactile sensing, hardware phase
Narrative bias	Worldbuilding preference is mistaken for empirical support	Separate evidence status and canon status in every artifact
Selection bias	Only successful evolutionary lineages are retained	Archive failures and report complete run manifests
Multiple testing	Many tasks produce chance "wins"	Preregister primary contrasts; control false discoveries in exploratory analyses

Stop Conditions

- Numerical instability, collision tunneling, or unbounded energy invalidates a run.
- A morphology is infeasible if it violates the resource envelope or relies on hidden support from the arm, table, or renderer.
- A controller is rejected if it exploits a simulator bug, nonphysical penetration, inaccessible state information, or evaluation leakage.
- A media result is rejected if task semantics are wrong even when the hand looks plausible.
- No biological claim advances beyond simulation without independent evidence and appropriate ethics review.

Phased Research Program

Phase	Purpose	Principal deliverables	Exit gate
0. Charter and preregistration	Freeze terminology, scope, budgets, metrics, governance	Protocol v1.1; data dictionary; threat model	Stakeholder and safety review
1. Infrastructure validation	Build generator, physics, tasks, replay, and baseline hands	Deterministic harness; unit tests; known-hand benchmarks	Reproducible baseline results
2. Pilot search	Estimate variance and detect design pathologies	Pilot archive; compute profile; revised parameter bounds	Stable metrics and feasible search
3. Confirmatory evolution	Run matched three-versus-five and full digit-count cohorts	Frozen candidate archive; blinded test results	Primary hypotheses adjudicated
4. Production validation	Run rigging and generative image/video comparisons	Scene corpus; QA logs; production-cost model	Independent evaluator agreement
5. Robustness and novelty	Test unseen objects, perturbations, engine transfer	Generalization report; failure taxonomy	No critical simulator exploit
6. Hardware feasibility	Fabricate a small set of Pareto candidates if warranted	Bench prototypes; calibration; sim-to-real comparison	Safety and value case approved
7. Application pilots	Test interfaces, tools, virtual species, or products	Entity-specific pilot packages	Separate authorization per entity
8. Canon/strategy review	Decide how findings may inform fictional or organizational systems	Evidence-status matrix; ratification packet	Human authority decision

Indicative Schedule

Window	Activity
Weeks 1–4	Protocol, requirements, terminology, asset licensing, threat model, benchmark selection
Weeks 5–12	Morphology generator, baseline hands, task suite, deterministic replay, pilot rendering scenes
Months 4–6	Pilot optimization, variance estimation, parameter correction, preregistration
Months 7–12	Confirmatory evolution and blinded evaluation
Months 10–14	Parallel generative-media and animation study
Months 13–18	Robustness, second-engine replication, and optional hardware prototype
After month 18	Entity-specific applications and canon review if authorized

The schedule is a planning baseline, not a promise. Compute requirements depend strongly on physics fidelity, controller algorithm, object diversity, and number of morphology evaluations. Pilot profiling must precede any budget commitment.

Research Team

- Program lead: scope, preregistration, stakeholder coordination, and decision gates.
- Morphology/robotics lead: kinematics, actuation, contact models, and manufacturability.
- Machine-learning lead: controller learning, co-design, evaluation isolation, and reproducibility.
- Biomechanics/evolution advisor: anatomical plausibility and limits on evolutionary interpretation.
- Graphics/animation lead: rigging, retargeting, mesh validity, and production-cost instrumentation.
- Generative-media lead: conditioning, fine-tuning, temporal evaluation, and model-card documentation.
- Statistician/methodologist: power simulation, hierarchical analysis, equivalence margins, and multiplicity.
- Data/governance steward: provenance, permissions, classifications, retention, and release review.
- Independent red team: simulator exploits, benchmark bias, and adversarial falsification.
- Canon/continuity reviewer: prevents unratified research from silently becoming universe canon.

Data, Governance, and Ethics

Evidence Status

Label	Meaning	Permitted use
Observed	Directly measured in a logged simulation or approved hardware test	Analysis within stated validity limits
Derived	Computed from observed data using documented code	Analysis with method and uncertainty attached
Modeled	Output of a calibrated model or simulator	Scenario and engineering inference, not direct reality claim
Hypothetical	Counterfactual morphology, species, universe, or technology	Simulation, foresight, and worldbuilding
Speculative	Interpretation not yet operationally tested	Idea generation only
Canon candidate	Worldbuilding proposition submitted for human ratification	No canonical use until approved
Operationally approved	Separately authorized application	Use only within the authorization scope

The project’s existing information-governance posture distinguishes admission from belief, retention, sharing, and action. This study follows that separation: a generated result may enter governed consideration without becoming true, canonical, shareable, or operationally authorized. Hypothetical representation is not assertion.

Ethical Limits

- No human or animal experimentation is included in the initial program.
- No inference should stigmatize people with limb differences or frame human anatomy as defective.
- If human motion-capture, clinical data, or artist performance data are later collected, obtain informed consent and define retention, secondary use, and deletion procedures.
- Accessibility research must involve affected users and must not assume that simplified morphology equals impaired capability.

- Hardware prototypes require force, pinch, electrical, and fail-safe risk assessment before human proximity testing.
- Generative training assets require lawful provenance, licensing, and identity/privacy controls.
- Public releases must include model limitations, task distributions, simulator assumptions, and negative results.

Run Manifest

Field group	Required fields
Identity	Run ID, project version, timestamp, responsible operator
Code	Repository commit, container hash, dependencies, engine version
Morphology	Genome, generated asset hash, mass, dimensions, DoF, actuators, sensors
Controller	Algorithm, architecture, parameters, initialization, checkpoint hash
Training	Task distribution, seed, steps, curriculum, compute, stopped/failed status
Evaluation	Frozen suite version, hidden-test token, objects, perturbations, metrics
Media	Prompt, references, mesh/keypoints, model version, seed, edits, repair time
Governance	Data class, access tier, license, release status, human approvals

Applications by Entity

The named organizations and universes are treated according to the limited definitions available in current project materials. OAL is identifiable as Omniversal Archive / Architect Labs. ONE INDUSTRIES and LEWIS CORP are not sufficiently defined for binding recommendations, so their applications below are provisional capability pathways. Any real-world or personal-data role for LEWIS CORP should default to narrower handling and stronger review.

OAL

- Establish TAMEL as a governed simulation branch preserving hypotheses, morphology lineages, failed designs, and uncertainty.
- Create an embodiment atlas linking hand morphology to tools, interfaces, gestures, arithmetic conventions, labor, and material culture.
- Use the quality-diversity archive as a scenario library rather than selecting one “correct” species design.
- Attach evidence labels to every claim and prevent modeled or speculative outputs from being mistaken for observed facts.
- Provide reusable benchmark worlds for embodied AI, character continuity, counterfactual evolution, and procedural civilization design.
- Preserve dissenting evaluations from independent AI and human reviewers before any ratification decision.

ONE INDUSTRIES

- Evaluate compact three-digit robotic end effectors for task-specific automation, hazardous environments, maintenance, and remote manipulation.
- Prototype co-designed tools, fasteners, grippers, controls, and chorded operator interfaces.
- Compare reduced actuator count and maintenance burden against losses in redundancy and generality.
- Use the production workstream to define asset standards for industrial digital twins and training simulators.
- Defer product, safety, IP, and manufacturing decisions until the organization's formal mandate and authority structure are defined.

LEWIS CORP

- Assess real-world-adjacent applications in assistive technology, custom prosthetic control, ergonomics, accessibility, and human-machine interfaces.
- Explore simulation services, digital prototyping, virtual training, and adaptive input devices without assuming clinical efficacy.
- Apply stricter privacy, consent, bias, safety, and evidence thresholds to any human data or medical-adjacent claim.
- Keep fictional-universe results separate from product claims and regulatory evidence.
- Commission independent review before any deployment involving people, workplaces, or health-related interpretation.

Simulated Environments

- Reduce character-hand topology and contact complexity where the creative design already supports tridactyl anatomy.
- Use a canonical three-digit skeleton and keypoint standard to improve cross-scene consistency.
- Benchmark typing, instrument performance, tool use, climbing, combat, caregiving, and construction as contact-grounded actions rather than decorative animation.
- Generate topology-aware tools and architecture so virtual civilizations are designed around their bodies instead of merely reskinned from human assets.
- Use morphology-conditioned controllers to populate diverse agents without hand-authoring every motion.

THE ONE UNIVERSE

Pending canonical definitions, treat THE ONE UNIVERSE as the primary high-coherence branch. Only validated simulation findings should constrain its anatomy and technology. A tridactyl species should be admitted as a candidate design only after its ecological niche, development, tools, gestures, injuries, interfaces, and cultural consequences are internally consistent.

- Adopt a reference anatomy sheet, skeletal standard, permitted variation range, and contact behavior.
- Derive industrial design from the hand: handles, locks, controls, instruments, weapons, medical tools, and writing systems.
- Use the simulation archive to prevent scene-to-scene drift in dexterity and strength.
- Separate “possible under the model” from “canonically present.”

THE MINUS ONE UNIVERSE

Pending canonical definitions, treat THE MINUS ONE UNIVERSE as a counterfactual and stress-test branch rather than a repository of inferior copies. It can explore constraint inversion: missing opposition, asymmetric digit counts, altered friction, hostile tool inheritance, unreliable sensing, reduced gravity, or unstable topology. The purpose is to reveal hidden assumptions in the primary branch and generate adversarial cases for the laboratory.

- Test whether three-digit success depends on friendly tools or environmental constants.
- Explore maladaptation, injury, developmental divergence, and incompatible inherited technology.
- Use failures to improve THE ONE UNIVERSE’s consistency without automatically transferring lore.
- Maintain explicit branch provenance so counterfactual findings cannot contaminate primary canon.

Real-World Implications

Domain	Potential value	Important limitation
Robotics	Task-specific hands with fewer components and high manipulation capacity	No guarantee of general human-tool compatibility
Prosthetics	Explore control-efficient terminal devices and individualized morphology	Clinical benefit requires user-centered trials and regulation
Accessibility	Alternative chorded, pressure, or multidirectional input systems	Must be designed with users, not imposed from simulation
Manufacturing	Lower actuator, wiring, maintenance, and collision complexity	Power, redundancy, and object diversity may suffer
Teleoperation	Compact controls mapped to optimized effectors	Embodiment mismatch may increase training burden
AI embodiment	Study how morphology shapes policy learning and world models	Results depend on task distribution and simulator assumptions
Generative media	Fewer joints and occlusions may improve topology consistency	Human-trained priors may resist novel anatomy without conditioning
Education/worldbuilding	Concrete demonstration of embodied cognition and coevolution	Speculation must remain labeled
Interface design	Chorded and continuous controls may outperform key arrays in niches	Learnability and accessibility must be measured
Safety engineering	Failure-tolerant morphology can be optimized explicitly	Three digits have less natural redundancy unless compensated

Likely Outcome Patterns

Scenario	Interpretation	Decision
Three dominates precision; five dominates power/redundancy	Expected specialist tradeoff	Develop task-specific three-digit systems; avoid universal claim
Three and five are equivalent under matched budgets	Digit count is less important than architecture and control	Select on cost, production, maintenance, or aesthetics
Five dominates most tasks	Additional contacts remain broadly valuable	Retain three for stylized media or narrow tools only
Three wins only with extra DoF/sensing	The gain comes from resource allocation, not count alone	Compare equal capability cost before adoption
Three wins only with coevolved tools	Anatomy–technology fit drives performance	Use in closed ecosystems; expect compatibility limits
General generators fail but mesh-guided systems succeed	Learned prior, not geometric impossibility, causes errors	Use structural conditioning and canonical rigs
Results reverse across physics engines	Contact-model dependence is too high	Delay conclusion; calibrate and replicate
One morphology dominates everything	Potential benchmark leakage or resource imbalance	Red-team before accepting an unusually universal result

Decision Gates

Gate	Question	Pass evidence
G0: Scope	Are terminology, authority, and non-objectives clear?	Approved protocol and evidence labels
G1: Infrastructure	Can runs be reproduced and replayed?	Hash-matched metrics and deterministic tests
G2: Fairness	Are resource and training budgets genuinely matched?	Independent accounting and ablation review
G3: Scientific signal	Does three show a robust, practically meaningful advantage anywhere?	Blinded held-out result with uncertainty
G4: Generalization	Does the result survive novel tasks and perturbations?	Bounded performance degradation
G5: Production value	Does three reduce total media cost or improve coherence?	Measured labor and error reduction
G6: Hardware value	Does a physical prototype preserve the advantage?	Calibrated bench replication
G7: Application	Is an entity-specific use justified and governed?	Safety, privacy, legal, and stakeholder authorization
G8: Canon	Should findings influence universe canon?	Separate human ratification; no automatic promotion

Required Deliverables

- Preregistered protocol and frozen confirmatory task suite.
- Parameterized 2–6 digit morphology library and validation tests.
- Baseline and co-designed controllers with learning curves.
- Pareto and MAP-Elites morphology archives, including failures.
- Blinded evaluation report with uncertainty and task-family breakdown.
- Rendered scene corpus and generative-media comparison dataset.
- Production labor, runtime, repair, and coherence report.
- Second-engine replication and sim-to-real feasibility report.
- Entity-specific application briefs with governance and authorization requirements.
- Canon-status packet clearly separating evidence, inference, speculation, and approved lore.

Appendix A — Minimum Experiment Matrix

Experiment	Digit counts	Optimization	Task ecology	Primary purpose
A1 Topology sweep	2, 3, 4, 5, 6	Fixed controller family; matched DoF	Mixed	Digit-count landscape
A2 Resource match	3 vs 5	Matched actuators, mass, energy, policy size	Mixed	Fair primary contrast
A3 Unconstrained best	2–6	Full co-design	Mixed	Best achievable per topology
A4 Precision niche	3 vs 5	Co-design	Precision-heavy	Test hypothesized tridactyl advantage
A5 Power niche	3 vs 5	Co-design	Power/robustness-heavy	Test expected disadvantage
A6 Tool inheritance	3 vs 5	Fixed versus coevolved tools	Interface/tool	Anatomy–technology coupling
A7 Failure tolerance	3 vs 5	Frozen elites	Digit/joint/sensor disable	Redundancy
A8 Engine transfer	Top Pareto designs	Frozen controllers	Second physics engine	Model dependence
A9 Media rigs	3 vs 5	Matched quality	Eight scene types	Animation cost/coherence
A10 Generative media	3 vs 5	Prompt, pose, mesh, fine-tune	Eight scene types	Prior versus structural complexity

Appendix B — Example Fitness Vector

Maintain a vector rather than one scalar:

$$F = [\text{task correctness, throughput, dexterous workspace, robustness, power, efficiency, generalization, redundancy, morphology simplicity, control simplicity, production coherence}]$$

If a scalar is temporarily needed for selection, use preregistered normalized terms and publish every component. Run alternative weight sets and show whether the ranking changes. No conclusion that depends on a single convenient weighting should be called general.

Appendix C — Evolution Pseudocode

```
initialize feasible morphology archive across digit counts 2..6
for generation in 1..G:
  select parents from Pareto and quality-diversity archives
  mutate morphology, sensing, compliance, and controller priors
  validate physical and resource constraints
  for each feasible offspring:
    train controller using the frozen curriculum and budget
    evaluate training, held-out, perturbation, and media tasks
    record raw trajectories, failures, energy, and provenance
    update Pareto archive and descriptor niche if improved
  periodically re-evaluate elites with new seeds and harder randomization
  freeze submissions before blinded confirmatory evaluation
  publish complete archive, uncertainty, ablations, and negative results
```

Appendix D — Data Dictionary

Variable	Type	Unit/example	Notes
digit_count	integer	3	Total digits, including opposable digit
total_dof	integer	12	All controllable hand DoF; wrist reported separately
actuator_count	integer	9	Independent powered channels
hand_mass	float	kg	Includes actuators, sensors, and palm
opposition_volume	float	m ³ or normalized	Reachable opposing-contact workspace
individuation	float	0–1	Observed independence, not command independence
task_success	binary/float	0/1 or graded	Defined by each task manifest
pose_rmse	float	mm/degrees	Object target error
disturbance_margin	float	N/Nm	Maximum retained external wrench
energy	float	J	Episode mechanical/electrical estimate
training_steps	integer	environment steps	Reported with wall-clock and hardware
contact_penetration	float	mm	Maximum and time-integrated
digit_validity	float	0–1	Species-specific anatomy validator
temporal_jitter	float	normalized	Residual unintended keypoint motion
repair_minutes	float	minutes	Human active time per accepted shot
evidence_status	category	observed/modeled/etc.	Governance label

Appendix E — Pre-registration Checklist

- Primary hypotheses and contrasts identified.
- Three digits defined as three total digits, including the opposable digit.
- Primary task families and practical equivalence margins frozen.
- Resource budgets and accounting rules frozen.
- Morphology feasibility constraints published.
- Controller families, training steps, stopping rules, and seeds specified.
- Confirmatory objects and perturbations isolated from developers.
- Exclusion criteria, simulator exploit policy, and failed-run handling defined.
- Statistical model and multiplicity strategy frozen.
- Production scenes, acceptance standards, and repair-time logging protocol frozen.
- Evidence labels, access controls, licensing, and release policy approved.
- Human authority for application and canon decisions identified.

Appendix F — Research Risk Register

Risk	Likelihood	Impact	Mitigation
Compute escalation from nested learning/evolution	High	High	Pilot profiling, surrogate models, early stopping with bias checks, parallelism
Contact simulation instability	Medium	High	Engine calibration, small time steps, penetration gates, second-engine replication
Benchmark favors desired answer	Medium	High	Independent task review, balanced ecologies, blinded testing
Novel anatomy validator unavailable	High	Medium	Build species-specific mesh/keypoint ground truth
General model reverts to five digits	High	Medium	Explicit skeleton/mesh conditioning and balanced fine-tuning
Result fails on hardware	Medium	High	Domain randomization, tactile sensing, calibrated prototypes
Fiction and evidence become conflated	Medium	High	Mandatory evidence/canon labels and human ratification
Human-data privacy issue in later phases	Low initially	High	No human data in base phase; consent and narrower controls later
Three-digit design stigmatizes limb difference	Low/Medium	High	Inclusive language, advisors, user-centered framing
Proprietary assets prevent reproducibility	Medium	Medium	Prefer open assets; publish substitutes and hashes

Appendix G — Interpretation Rules

- › A three-digit win on one task means local superiority, not global superiority.
- › A simulated evolutionary outcome means success under encoded selection pressures, not historical inevitability.
- › A lower joint count means lower kinematic complexity only if compensatory joints, sensing, and palm mechanisms are included in the accounting.
- › A faster rendered frame does not establish lower production cost; artist and QA time must be measured.
- › A plausible-looking hand does not establish correct task contact.
- › A human-hand detector score does not establish validity for a nonhuman anatomy.
- › A coevolved tool advantage may be valuable in a closed ecosystem but does not imply compatibility with the human world.
- › A worldbuilding application remains speculative until separately ratified.
- › Negative results are program assets because they map the limits of the design space.
- › The strongest conclusion available from this program is a conditional design map: which morphology works best, for which tasks, under which constraints.

Closing Research Position

The proposal survives critical review when stated narrowly and tested fairly. Three total digits can form a complete manipulation architecture, and fewer digits may reduce some control, rigging, and coherence burdens. Yet the saved complexity must be reinvested in opposition, independence, joint range, palm behavior, sensing, and task-appropriate tools. Five digits retain credible advantages in contact area, redundancy, power, span, and simultaneous object management.

TAMEL-1 is designed to discover that boundary instead of assuming it. Its success is not defined by proving three digits “better.” Success is a reproducible, adversarially tested map of morphology–control–environment tradeoffs that can support engineering, virtual production, embodied-AI research, and internally governed universe design without confusing simulation with biological fact or canon.